

Lipidomic analysis and dengue virus infection

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Collaborator
Dorothee MISSE

IRD 911 Avenue Agropolis 34394 Montpellier

Report author
Cyrielle CLEMENT
MTH-MetaToul Lipidomique

People involved in the project

Platform staff

Cyrielle CLEMENT

Anaëlle DURBEC

Julia SOULLIER

Océane DELOS

Nancy GEOFFRE

Table of contents

Summary of main results	3
Context	4
Samples	5
Validation	Erreur ! Signet non défini.
Results	6
Experimental details	7
METABOHUB, the National infrastructure in metabolomics & fluxomics	10

Summary of main results

Your lipidomic assays went well, all our controls (instrument status, internal standards, blank) were correct.

For the fatty acids analysis, a mix-up between samples 67 and 68 was identified, requiring a re-extraction.

Context

This project initiated by MetaToul platform Dengue fever virus (DENV) is the most widespread arbovirus and a major threat to public health in tropical and subtropical areas of the world. Dengue is arguably the world's most important arboviral disease, with around 390 million infections per year, of which almost 100 million are clinically apparent. While the majority of people infected with DENV recover rapidly, almost 500,000 people a year develop a severe form of the disease, classified as either dengue hemorrhagic fever (DHF) or dengue shock syndrome (DSS). Characterized by increased vascular permeability, hypovolemia and dysregulated blood coagulation, DHF/DSS has a mortality rate of 2.5%1-3. Epidemics of dengue fever (DF) and dengue hemorrhagic fever/dengue shock syndrome (DHF/DSS) have appeared throughout the tropical and subtropical world, with devastating public health consequences for diagnosis and patient care. If the disease is presented early after onset, it is clinically impossible to differentiate between patients infected with DENV, who will experience a mild episode of dengue disease, and those who will progress to potentially fatal DHF/DSS. Moreover, many questions remain unanswered about the molecular mechanisms underlying the infection, and there is currently no specific treatment or vaccine effective against the disease.

Samples

Nature of samples

- Origin(s): Plasma

Received on: 24/09/25

Extracted on: protocol oxylipins and global: 10/11/24 - 11/11/24 and FAME: 09/01/25 and 15/01/25

Analyzed on: oxylipins: 21/10/24; globale : 28/10/24 and FAME: 10/01/25

Sample list:

Analyse number	Sample	Analyse num	Sample	Analyse num	Sample	Analyse num	Sample	Analyse num	Sample
11	BS-064D0	49	JV-035D60	102	KT-347D0	161	KT-525D60	43	KT-716D60
33	BS-064D03	124	JV-048D0	44	KT-347D03	163	KT-537D0	40	KT-718D0
136	BS-064D10	78	JV-048D03	36	KT-347D10	150	KT-537D03	50	KT-718D03
108	BS-064D60	20	JV-048D10	142	KT-347D60	95	KT-537D10	86	KT-718D10
68	BS-082D0	152	JV-071D0	105	KT-352D0	34	KT-537D60	151	KT-718D60
64	BS-082D03	116	JV-071D03	94	KT-352D03	85	KT-538D0	51	KT-723D0
41	BS-082D10	30	JV-071D60	98	KT-352D10	87	KT-538D03	109	KT-723D03
10	BS-082D60	5	JV-138D0	145	KT-352D60	47	KT-538D10	7	KT-723D10
154	BS-336D0	76	JV-138D03	170	KT-412D0	110	KT-538D60	13	KT-723D60
24	BS-336D03	38	JV-138D10	53	KT-412D03	132	KT-539D0	113	KT-771D0
12	BS-336D10	37	JV-138D60	57	KT-412D10	133	KT-539D03	106	KT-771D03
103	BS-336D60	157	JV-148D0	3	KT-412D60	162	KT-539D10	25	KT-771D10
137	BS-346D0	59	JV-148D03	56	KT-417D0	125	KT-539D60	122	KT-771D60
19	BS-346D03	22	JV-148D10	159	KT-417D03	147	KT-565D0	45	KT-805D0
101	BS-346D10	17	JV-148D60	123	KT-417D10	139	KT-565D03	127	KT-805D03
120	BS-346D60	16	JV-157D0	75	KT-417D60	128	KT-565D10	77	KT-805D10
21	BS-351D0	81	JV-157D03	168	KT-434D0	27	KT-612D0	88	KT-805D60
46	BS-351D03	146	JV-157D10	9	KT-434D03	148	KT-612D03	126	KT-880D0
74	BS-351D10	69	JV-157D60	63	KT-434D10	14	KT-612D10	65	KT-880D03
129	BS-351D60	111	KT-193D0	158	KT-434D60	90	KT-612D60	155	KT-880D10
2	BS-364D0	92	KT-193D03	42	KT-445D0	18	KT-663D0	167	KT-880D60
144	BS-364D03	31	KT-193D10	71	KT-445D03	130	KT-663D03	91	KT-926D0
107	BS-364D10	156	KT-193D60	112	KT-445D10	160	KT-663D10	29	KT-927D03
83	BS-364D60	32	KT-247D0	62	KT-445D60	93	KT-663D60	141	KT-928D10
35	BS-377D0	39	KT-247D03	58	KT-515D0	73	KT-695D0	23	KT-929D60
166	BS-377D03	114	KT-247D10	104	KT-515D03	26	KT-695D03	82	KT-974D0
72	BS-377D10	66	KT-247D60	52	KT-515D10	131	KT-695D10	4	KT-974D03
55	BS-377D60	70	KT-312D0	115	KT-515D60	140	KT-695D60	149	KT-974D10
134	BS-671D0	153	KT-312D03	143	KT-522D0	79	KT-705D0	172	KT-974D60
6	BS-671D03	1	KT-312D10	118	KT-522D03	15	KT-705D03		
171	BS-671D10	28	KT-312D60	138	KT-522D10	67	KT-705D10		
80	BS-671D60	61	KT-313D0	97	KT-522D60	8	KT-705D60		
96	JV-035D0	164	KT-313D03	119	KT-525D0	100	KT-716D0		
169	JV-035D03	54	KT-313D10	121	KT-525D03	99	KT-716D03		
165	JV-035D10	135	KT-313D60	117	KT-525D10	84	KT-716D10		

Results

→ Oxylipins results are synthesized in excel file: 241009DMI_oxylipins_results

Absolute quantification was performed. Values are expressed in ng/ μ L of plasma.

→ Globale results are synthesized in excel file: MISSE_Global_Lipidomic_results

Relative quantification was performed. Values are expressed in (metabolite area/ISTD area)/ μ L of plasma.

→ FAME results are synthesized in excel file: 241109DMI_Rendu_AGT

Relative quantification was performed. Values are expressed in (metabolite area/ISTD area)/ μ L of plasma.

NB: If you use these datas for communication (poster, oral communication, scientific article), please thanks the lipidomic facility core as mentionned:" MetaToul-Lipidomique (I2MC, Inserm, Toulouse, France), MetaboHUB-ANR-11-INBS-0010".

Experimental details

Extraction and analysis, Total fatty acids

Scope & application

Quantitative analysis of conventional fatty acids: (c10:0, c12:0, c14:0, c15:0, c16:0, c17:0, c18:0, c20:0, c22:0, c23:0, c24:0, c14:1w5, c15:1, c16:1w7, c18:1w9, c18:1w7, c20:1w9, c22:1w9, c24:1w9, c18:2w6, c18:3w6, c18:3w3, c20:2w6, c20:3w3, c20:3w6, c20:4w6, c20:5w3, c22:2w6, c22:6w3, c22:4w6) with internal calibration (internal standard: TG17).

This method allows a relative quantification of the molecules.

Sample preparation:

Samples were reworked with 3x 100 μ L ethyl acetate after analysis in total and then fractions were hydrolysed in KOH (0.5 M in methanol, 1mL) at 55°C for 30 min, and transmethylated in boron trifluoride methanol solution 14% (SIGMA, 1 mL) and heptane (1 mL) at 80°C for 1h. After addition of water (1 mL), FAMES were extracted with heptane (2mL), centrifugate 1min at 2500RPM and evaporated to dryness and dissolved in ethyl acetate (20 μ L).

Total fatty acids' fractions were trans methylated in boron trifluoride methanol solution 14% (SIGMA, 1 mL) and heptane (1 mL) ambient temperature during 5min. After addition of water (1 mL) to the crude, FAMES were extracted with heptane (2mL), centrifugate 1min at 2500RPM and evaporated to dryness and dissolved in 15 μ L of ethyl acetate.

Analysis:

FAME (1 μ L) were analyzed by gas chromatography equipped with a FID (Lillington, J.M., Trafford, D.J., and Makin, H.L. 1981, Clin Chim Acta 111:91-98) on a GC Trace 1600 system using a Famewax RESTEK fused silica capillary columns (30 m x 0.32 mm i.d, 0.25 μ m film thickness). Oven temperature was programmed from 100°C to 250°C at a rate of 6°C/min and the carrier gas was hydrogen (1.5mL/min). The injector and the detector were at 220°C and 230°C respectively.

Experimental details

Extraction and analysis of oxylipins

1. Scope & application

Quantitative analysis of the following classes: prostaglandins, thromboxans, leukotriens, lipoxins, resolvins, HODE, HETE, HDoHE, EET, through internal calibration.

Sample preparation:

260 μ L of cold methanol and 40 μ L of internal standard (Deuterium labeled compounds) were added to 120 μ L (or less) of plasma. After centrifugation at 5000 g for 15 min at 4°C, supernatants were transferred into 2 mL 96-well deep plates and diluted in H₂O to 2 mL. Samples were then submitted to solid phase extraction (SPE) using OASIS HLB 96-well plate (30 mg/well, Waters) pretreated with MeOH (1 mL) and equilibrated with 10% MeOH (1 mL). After sample application, extraction plate was washed with 10% MeOH (1 mL). After drying under aspiration, lipids mediators were eluted with 1 mL of MeOH. Prior to LC-MS/MS analysis, samples were evaporated under nitrogen gas and reconstituted in 20 μ L on MeOH.

Analysis:

Briefly, lipid mediators were separated on a Kinetex XB-C18 column (2.1 mm, 100 mm, 1.8 μ m) (Agilent Technologies) using Nexera Mikros HPLC system (Shimadzu) coupled to a ESI-triple quadruple 8060 mass spectrometer (Shimadzu). Data were acquired in Dynamic Multiple Reaction Monitoring (DMRM) mode with optimized conditions (ion optics and collision energy). Peak detection, integration and quantitative analysis were done using LabSolution analysis software (Shimadzu) based on calibration lines built with commercially available oxylipins standards (Cayman Chemicals).

Ref : Characterization of the Oxylipin Pattern and Other Fatty Acid Oxidation Products in Freshly Pressed and Stored Plant Oils, Koch E. et al, J. of Agri. Food Chem., 2022

Experimental details

Extraction and analysis of global lipidomic

Scope & application

Quantitative analysis of the following classes: Glycerolipids (TG, DG); Cholesteryl ester (CE); Sphingolipids (Ceramide, SM, Mono-hexosylCeramide, Di-hexosylceramide); phospholipids (PC, LPC, PE, LPE, PI, PS, PG); Free fatty acids (FA), Acyl carnitines (CAR) with internal standard. The analysis will provide the length of the fatty acid chain and the number of unsaturation but not their positions.

Sample preparation

10 μ L of plasma were added to a glass tube containing 10 μ L of a mixture of internal standards. Lipids were extracted according to a modified Folch extraction (Seyer et Al., Metabolomics, DOI: <https://doi.org/10.1007/s11306-016-1023-8>, Folch et Al., J. Biol. Chem). A volume sufficient for 100 μ L of ultrapure water is added along with 490 μ L of a mixture of CHCl₃/MeOH (1:1 v/v). The sample was vortexed during 15 minutes. 75 μ L of ultra pure H₂O is added before vortexing again for one minute and then centrifuging at 2500 rpm for 15 minutes. The upper and the lower phase was then transferred to a glass tube and evaporated under nitrogen and reconstituted in 100 μ L of a 65:35:5 (v/v/v) MeOH/IPA/H₂O mixture.

Analysis

SFC-MS analysis were performed as described (Le Faouder et Al., Metabolites, DOI: 10.3390/metabo11050305). Briefly, lipid mediators were separated on a ACQUITY UPC² Torus diethylamine column (100x3.0 mm, 1.7 μ m, Waters) using an Ultra-Performance Convergence Chromatography (UPC²) coupled on-line to an Xevo G2-XS time of flight (Qtof) (Waters, Milford, MA, USA), equipped with an ESI source. The analyses were performed in MS full scan in centroid mode from 50 to 1500 Da, with dynamic range extended (DRE) activated.

Peak integration and quantitative analysis were done using MzMine4.

METABOHUB, the National infrastructure in metabolomics & fluxomics

MetaboHUB is the French Infrastructure in Metabolomics & Fluxomics created in 2013 in the framework of the Program “Investissements d’Avenir” (Investment for the Future) launched by the French Ministry of Research, Higher Education and Innovation (MESRI) and the National Agency for Research (ANR).

MetaboHUB addresses key critical priorities in metabolomics, fluxomics and bioinformatics to (i) provide high throughput quantitative analytical technologies for biochemical phenotyping of large variety of biological samples, (ii) identify metabolites and annotate metabolomes, (iii) develop broadband flux measurements, (iv) provide access to high value-added services to national scientific community and industry players, (v) attract and train a new generation of scientists and users.

